Slipping rope — Solution

W24 (2024) — English translation

A1 **1.50 pts**

Determine the speed v and acceleration a of the left end of the rope at the moment when the left end is lowered by the amount x. Express your answer using g, L_1, L, α and x.

First solution: The length of the part of the rope hanging on the right is $L_2 = L_1 + L\sin\alpha$, and the total length of the rope is:

$$l = 2L_1 + L(1 + \sin\alpha).$$

Let us use законом сохранения механической энергии:

$$\frac{mv^2}{2} = -\Delta W_p,$$

where $m = \lambda l$ – mass верёвки, а ΔW_p – изменение её potentialьной энергии в поле тяжести. Fromменение potentialьной энергии верёвки в поле тяжести equals изменению potentialьной энергии участка верёвки длиной x, координата x_C центра масс которого изменилась от значения $x_{C_1} = -x/2$ до $x_{C_2} = x/2$. Hence we have:

$$\Delta W_p = -\lambda x \cdot g \cdot (x_{C_2} - x_{C_1}) = -\lambda g x^2.$$

Substituting ΔW_p в conservation of mechanical energy, we get:

$$\frac{mv^2}{2} = \frac{\lambda l v^2}{2} = \lambda g x^2 \Rightarrow v = x\sqrt{\frac{2g}{l}},$$

from:

$$l = 2L_1 + L(1 + \sin\alpha).$$

Let's use the law of conservation of mechanical energy:

$$\frac{mv^2}{2} = -\Delta W_p,$$

where $m = \lambda l$ is the mass of the rope, and ΔW_p is the change in its potential energy in the gravitational field.

The change in the potential energy of the rope in the gravitational field is equal to the change in the potential energy of a section of the rope of length x, the coordinate x_C of the center of mass of which has changed from the value $x_{C_1} = -x/2$ to $x_{C_2} = x/2$. From here we have:

$$\Delta W_p = -\lambda x \cdot g \cdot (x_{C_2} - x_{C_1}) = -\lambda g x^2.$$

Substituting ΔW_p into the law of conservation of mechanical energy, we obtain:

$$\frac{mv^2}{2} = \frac{\lambda l v^2}{2} = \lambda g x^2 \Rightarrow v = x\sqrt{\frac{2g}{l}},$$

where:

Since the speed $v(x)$ and the displacement x are related by direct proportionality:

$$a = v\sqrt{\frac{2g}{l}} = \frac{2gx}{l},$$

from where:

Second solution: Let's write down the equation of motion for each section of the rope:

$$\lambda(L_1 + x)a = \lambda(L_1 + x)g - T_1 \qquad \lambda La = T_1 - T_2 + \lambda gL\sin\alpha \qquad \lambda(L_2 - x)a = T_2 - \lambda(L_2 - x)g,$$

where T_1 and T_2 – силы натяжения верёвки вблfromand pulleyов 1 and 2 соответственно. Складывая уравнения, we get:

$$\lambda(L_1 + L_2 + L)a = \lambda g(L_1 + L\sin\alpha - L_2 + 2x),$$

from where

$$\lambda(L_1 + x)a = \lambda(L_1 + x)g - T_1 \qquad \lambda La = T_1 - T_2 + \lambda gL\sin\alpha \qquad \lambda(L_2 - x)a = T_2 - \lambda(L_2 - x)g,$$

where T_1 and T_2 are the rope tension forces near blocks 1 and 2, respectively.
Adding the equations, we get:

$$\lambda(L_1 + L_2 + L)a = \lambda g(L_1 + L\sin\alpha - L_2 + 2x),$$

where
Multiplying the expression for acceleration by speed, we get:

$$av = \frac{d(v^2/2)}{dt} = \frac{2gxv}{2L_1 + L(1+\sin\alpha)} = \frac{g}{2L_1 + L(1+\sin\alpha)}\frac{d(x^2)}{dt}$$

Integrating, we get:

$$\frac{v^2}{2} = \frac{gx^2}{2L_1 + L(1+\sin\alpha)},$$

from:

Answer: \textbfAnswer:

$$v = x\sqrt{\frac{2g}{2L_1 + L(1+\sin\alpha)}}.$$

A2 — 0.80 pts

Determine the amount of displacement of the left end of the rope x_1 at the moment in time when the tension of the section of the rope lying on the inclined plane is constant along the length of this section. Express your answer in terms of L_1, L and α. At what ratios L_1, L and α is the value x_1 less than the initial length of the right section of the rope?

Let us introduce the axis y, directed along the inclined plane from block 1 to block 2. Let us write in projection onto the axis y Newton's second law for a section of rope of length dy lying on an inclined plane:

$$dT + \lambda g\sin\alpha dy = \lambda a dy.$$

The quantity $dT = 0$ for/at $a(x_1) = g\sin\alpha$. Hence:

$$\frac{2gx_1}{l} = g\sin\alpha \Rightarrow x_1 = \frac{l\sin\alpha}{2},$$

from where:

Let's compare the value x_1 with the value L_2:

$$x_1 = L_1 \sin\alpha + \frac{L\sin\alpha(1+\sin\alpha)}{2} \qquad L_2 = L_1 + L\sin\alpha.$$

Coefficientы перед L_1 and L в выражении for L_2 больше соответствующих coefficientов в выражении for x_1, therefore:

Answer: \textbfAnswer:

$$x_1 = L_1 \sin\alpha + \frac{L\sin\alpha(1+\sin\alpha)}{2}.$$

A3 — 1.50 pts

Determine the maximum tension force of the rope at the moment of time when its left end has shifted by a distance x. Express your answer in terms of λ, g, L_1, L, α and x.

On vertical sections of the rope, the greatest tension force is achieved at the points of contact of the blocks, since the acceleration $a(x)$ is always less than g. Let us denote by T_1 and T_2 the tension force of the rope at the points of contact of blocks 1 and 2, respectively. From Newton's second law for the left vertical section of the rope we obtain:

$$\lambda(L_1+x)g - T_1 = \lambda(L_1+x)a \qquad T_2 - \lambda(L_2-x)g = \lambda(L_2-x)a,$$

hence:

$$T_1 = \lambda(L_1+x)(g-a) \qquad T_2 = \lambda(L_2-x)(g+a).$$

Проаналfromandруем наклонный учаccurrent верёвки. From решения parta A2 it follows:

$$\frac{dT}{dy} = \lambda(a - g\sin\alpha).$$

For/At $x \le x_1$ quantity dT/dy не положительна, а for/at $x \ge x_1$ – неотрицательна, hence it follows, что:

$$T_{max} = \begin{cases} T_1 & \text{при} \quad x \ge x_1 \\ T_2 & \text{при} \quad x \le x_1 \end{cases}$$

Substituting выражения for a, L_2 and x_1, we get:

Answer: \textbfAnswer:

$$T_{max} = \begin{cases} \dfrac{\lambda g(L_1+x)(2L_1+L(1+\sin\alpha)-2x)}{2L_1+L(1+\sin\alpha)} & \text{при} \quad x \ge L_1\sin\alpha + \dfrac{L\sin\alpha(1+\sin\alpha)}{2} \\ \dfrac{\lambda g(L_1+L\sin\alpha-x)(2L_1+L(1+\sin\alpha)+2x)}{2L_1+L(1+\sin\alpha)} & \text{при} \quad x \le L_1\sin\alpha + \dfrac{L\sin\alpha(1+\sin\alpha)}{2} \end{cases}$$

A4 — 2.50 pts

Determine the displacement values $x_{max(1)}$ and $x_{max(2)}$ of the left end of the rope, at which it begins to lose contact with blocks 1 and 2, respectively, assuming that contact with the other block is maintained. Express your answers using L_1, L and α.

Let us write down Newton's second law for a section of rope with mass $dm = \lambda dl$:

$$dm\vec{a} = d\vec{T} + dm\vec{g} + d\vec{N}.$$

At the moment of detachment $d\vec{N} = 0$, therefore:

$$dm\vec{a} = d\vec{T} + dm\vec{g}.$$

Let ρ – radius кривизны рассматриваемой поверхности, then $dl = \rho d\varphi$, where $d\varphi$ – angle поворота касательной к верёвке на рассматриваемом участке dl. Спроецируем equation движения на направление нормали, направленное к центру кривизны:

$$dma_n = \lambda\rho d\varphi \cdot \frac{v^2}{\rho} = \lambda v^2 d\varphi = T d\varphi + \lambda\rho d\varphi \cdot g_n.$$

where g_n – проекция ускорения свободного падения на направление нормали. Since radius pulleyов мал – condition отрыва the following:

$$T = \lambda v^2.$$

Временно передеnote величины $x_{max(1)}$ and $x_{max(2)}$ за x_1 and x_2 соответственно. Equation for x_1 the following:

$$\frac{\lambda g(L_1 + x_1)(2L_1 + L(1+\sin\alpha) - 2x_1)}{2L_1 + L(1+\sin\alpha)} = \frac{2\lambda g x_1^2}{2L_1 + L(1+\sin\alpha)}.$$

hence:

$$4x_1^2 - x_1 L(1+\sin\alpha) - L_1(2L_1 + L(1+\sin\alpha)) = 0.$$

У This quadratic equation has one positive root, therefore:

The equation for x_2 is:

$$\frac{\lambda g(L_1 + L\sin\alpha - x_2)(2L_1 + L(1+\sin\alpha) + 2x_2)}{2L_1 + L(1+\sin\alpha)} = \frac{2\lambda g x_2^2}{2L_1 + L(1+\sin\alpha)},$$

hence:

$$4x_2^2 + x_2 L(1-\sin\alpha) - (L_1 + L\sin\alpha)(2L_1 + L(1+\sin\alpha)) = 0.$$

This quadratic equation has one positive root, so:

Answer: \textbfAnswer:

$$x_{max(1)} = \frac{L(1+\sin\alpha)}{8} + \sqrt{\left(\frac{L(1+\sin\alpha)}{8}\right)^2 + \frac{L_1(2L_1 + L(1+\sin\alpha))}{4}}.$$

A5 — 1.20 pts

Calculate the speed of the rope at the moment of its separation from one of the blocks if the inclined plane forms angles $\alpha_1 = 30°$, $\alpha_2 = 45°$ and $\alpha_3 = 60°$ with the horizon.

For each of the angles, separation is achieved at x, equal to the smaller of the values $x_{max(1)}$ and $x_{max(2)}$ corresponding to a given angle. Let's calculate them:

$\alpha,^\circ$ $x_{max(1)}$, m $x_{max(2)}$, m 30 0.777 0.731 45 0.833 0.868 60 0.876 0.973

Thus, at angle α_1 the rope first loses contact with the second block, and at angles α_2 and α_3 – with the first. From here we find:

Answer: \textbfAnswer:

$$v_1 = 2.068 \text{ m/s} \qquad v_2 = 2.264 \text{ m/s} \qquad v_3 = 2.314 \text{ m/s}.$$

B1 — 0.80 pts

Determine the displacement value x_0 of the left end of the rope corresponding to the equilibrium position of the system. Express your answer in terms of m, λ, L_1, L, α, g, k and F_0.

Let's write down the equilibrium conditions for three sections of the rope:

$$T_1 = \lambda g(L_1 + x_0) + mg \qquad T_2 - T_1 = \lambda g L\sin\alpha \qquad T_2 = \lambda g(L_2 - x_0) + F_{\text{упр}},$$

hence:

$$F_{\text{упр}} = \lambda g(L\sin\alpha + L_1 + x_0 + x_0 - L_1 - L\sin\alpha) + mg = mg + 2\lambda g x_0.$$

Since $F_{\text{control}} = F_0 + kx_0$, we get:

Answer: \textbfAnswer:

$$x_0 = \frac{mg - F_0}{k - 2\lambda g}.$$

B2 — 0.70 pts

The equilibrium position from the previous paragraph exists if the spring stiffness exceeds some minimum value k_{min}. Identify k_{min}. Express your answer using m, λ, L_1, L, α, g, k and F_0. Further, assume everywhere that $k > k_{min}$.

The denominator of the result obtained in point B1 must be positive, so we have the limitation:

$$k > 2\lambda g.$$

The quantity x_0 also ограничена длиной L_2 правого вертикального участка верёвки, therefore:

$$x_0 = \frac{mg - F_0}{k - 2\lambda g} \leq L_1 + L\sin\alpha \Rightarrow k \geq 2\lambda g + \frac{mg - F_0}{L_1 + L\sin\alpha}.$$

The last limitation is more significant, so we have:

Answer: \textbfAnswer:

$$k_{min} = 2\lambda g + \frac{mg - F_0}{L_1 + L\sin\alpha}.$$

B3 — 0.80 pts

Find the change in the potential energy of the system ΔW_p when the left end of the rope is displaced by the amount x. Express your answer in terms of m, λ, L_1, L, α, g, k, F_0 and x.

For quantities $\Delta W_{p(\text{g})}$ and $\Delta W_{p(\text{в})}$ we have:

$$\Delta W_{p(\text{g})} = -mgx \qquad \Delta W_{p(\text{g})} = -\lambda g x^2.$$

Fromменение potentialьной энергии деформации пружины $\Delta W_{p(\text{def})}$ equals площади под графиком $F_{\text{control}}(\Delta l)$ and составляет:

$$\Delta W_{p(\text{деф})} = F_0 x + \frac{kx^2}{2}.$$

Finally:

Answer: \textbfAnswer:

$$\Delta W_p = \frac{(k - 2\lambda g)x^2}{2} - (mg - F_0)x.$$

B4 — 0.70 pts

If we take the potential energy W_p at the equilibrium position equal to zero, then it can be represented in the form

$$W_p = \frac{A(x - B)^2}{2}.$$

Determine A and B. Answerы вытimesandme via/through m, λ, L_1, L, α, g, k and F_0.

From the expression for ΔW_p we have:

$$W_p = \frac{(k - 2\lambda g)x^2}{2} - (mg - F_0)x + C = \frac{k - 2\lambda g}{2}\left(x - \frac{mg - F_0}{k - 2\lambda g}\right)^2 - \frac{(mg - F_0)^2}{2(k - 2\lambda g)} + C.$$

Olimitяя potentialьную энергию в положении equalsвесия equalsй нулю:

$$W_p = \frac{k-2\lambda g}{2}\left(x - \frac{mg-F_0}{k-2\lambda g}\right)^2.$$

Thus:

Answer: \textbfAnswer:

$$A = k - 2\lambda g \qquad B = \frac{mg-F_0}{k-2\lambda g}.$$

B5 **0.50 pts**

Determine the maximum speed of the load v_{max} and its displacement from the initial position x_{max} if the system is released without an initial speed. Express your answers using m, λ, L_1, L, α, g, k and F_0.

The maximum speed of the load is achieved at the point with coordinate x_0 and is determined from the law of conservation of mechanical energy:

$$\frac{(m+\lambda(2L_1+L(1+\sin\alpha))v_{max}^2}{2} = \frac{Ax_0^2}{2},$$

from where:

The value x_{max} is achieved at the moment the load stops, i.e. is determined from the condition:

$$\frac{A(x_{max}-x_0)^2}{2} = \frac{Ax_0^2}{2} \Rightarrow x_{max} = 2x_0,$$

from where:

Answer: \textbfAnswer:

$$v_{max} = \frac{mg-F_0}{k-2\lambda g}\sqrt{\frac{k-2\lambda g}{m+\lambda(2L_1+L(1+\sin\alpha)}}.$$

B6 **1.00 pts**

What minimum speed v_{min} must the weight have in the initial position for the right end of the thread to reach the level of block 2? Express your answer using m, λ, L_1, L, α, g, k and F_0.

The load must stop at the moment when it drops by the amount L_2. From the law of conservation of mechanical energy:

$$\frac{A(L_2-x_0)^2}{2} = \frac{Ax_0^2}{2} + \frac{(m+\lambda(2L_1+L(1+\sin\alpha))v_{min}^2}{2},$$

from:

Answer: \textbfAnswer:

$$v_{min} = \sqrt{\frac{k-2\lambda g}{m+\lambda(2L_1+L(1+\sin\alpha)}(L_1+L\sin\alpha)\left(L_1+L\sin\alpha-\frac{2(mg-F_0)}{k-2\lambda g}\right)}$$